

Unit 4 Handout: Discrete Random Variable

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Example 1: Back to the Basket!

Suppose, in a basket we have 10 balls. Seven of them are red, one is black, and the remaining are green. We are going to draw two balls from the basket at random **with replacement (WR)**. Let X = the number of red-ball draws = $\{0, 1, 2\}$.



(a) Find the probability mass function (pmf) $p(x) = P(X = x)$ for all possible values of X .

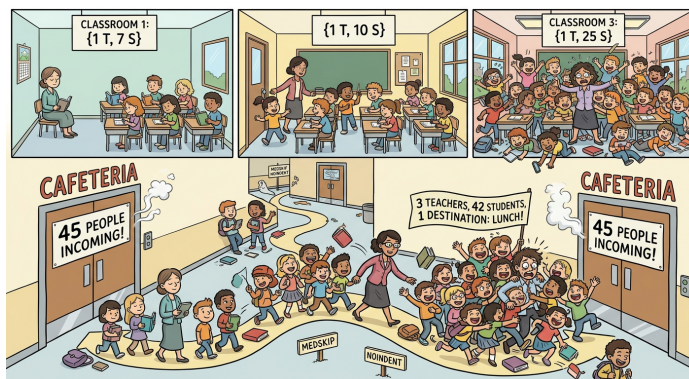
(b) Find the cumulative distribution function (CDF) $F(x) = P(X \leq x)$.

- (c) Now, suppose we repeatedly draw a ball **with replacement** until we observe the first black ball. Let X_B = the number of draws required to observe the first black ball so that $X_B = \{1, 2, 3, 4, \dots\}$.

Discuss this scenario as a **geometric series**.

- (d) Find $\mu_X = E(X)$, $\mu_{X_B} = E(X_B)$ and σ_X .

Example 2: Lunch Time!



Suppose that there are three first-grade classroom sections at Boulevard Elementary School:

- {1 teacher, 7 students}
- {1 teacher, 10 students}
- {1 teacher, 25 students}

At lunch time, all 3 teachers and all 42 students walk to the cafeteria.

Scenario 1: Select one student at random.

Ask the selected student how many students are in their class. Let

X = number of students in the selected student's class = {7, 10, 25}.

Scenario 2: Select one teacher at random.

Ask the selected teacher how many students are in their class. Let

Y = number of students in the selected teacher's class = {7, 10, 25}.

(a) Calculate $\mu_X = E(X)$ and $\mu_Y = E(Y)$.

(b) Find the variance and standard deviation for both X and Y

Example 3: Loaded Die!



Let X be the outcome of a loaded die: $X = \{1, 2, 3, 4, 5, 6\}$

The probability mass function is given below.

X	$P(X = x)$	$x - \mu_X$	$(x - \mu_X)^2$	x^2
1	0.1			
2	0.1			
3	0.2			
4	0.2			
5	0.1			
6	0.3			

(a) Find the population mean (μ_X) and population standard deviation σ_X .

(b) Let $Y = 2X + 3$. Using the propositions of expected value and variance, find $E(Y)$ and $V(Y)$.

(c) What if we have $n = 1000$ die rolls?

Example 4: Let's Play Roulette!



A roulette wheel is divided into 38 slices:

1–36: 18 red and 18 black

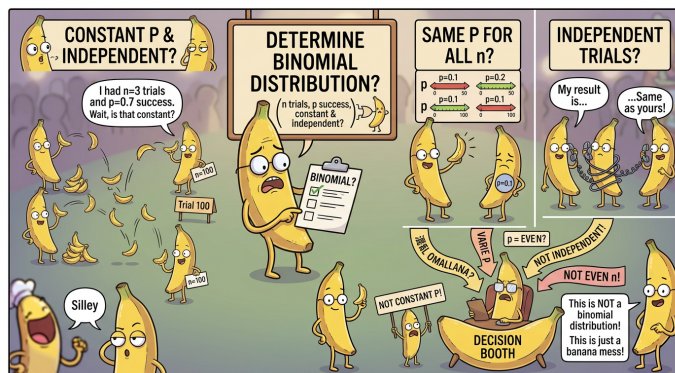
0 and 00: 2 green

Let, Y = profit from a \$1 bet on red.

In the long run, do you expect to **win money**, **lose money**,
or **break even**? Why?



Example 5: Binomial Distribution



Determine whether each of the following random variables follows a **binomial distribution**. Justify your answer.

- Suppose a fair coin is flipped 6 times. Let,
 X = number of tails observed in the 6 flips = $\{0, 1, 2, 3, 4, 5, 6\}$
- Suppose a fair coin is repeatedly flipped until a total of 3 tails have been observed. Let,
 Y = number of flips required to observe 3 tails = $\{3, 4, 5, \dots\}$
- A basket contains 10 balls: 7 red, 2 green, and 1 black. Four balls are drawn at random **without replacement (WOR)**. Let,
 Z = number of red balls drawn = $\{1, 2, 3, 4\}$
- Consider the same basket, but now the four balls are drawn **with replacement (WR)**. Let
 W = number of red balls drawn = $\{0, 1, 2, 3, 4\}$

Example 6: Fluorescent Light Bulbs

A company manufactures fluorescent light bulbs. Let $W = \{\text{a bulb is still working after 10,000 hours of use}\}$. Suppose, $P(W) = 0.86$, and assume that the lifetimes of the bulbs are independent.



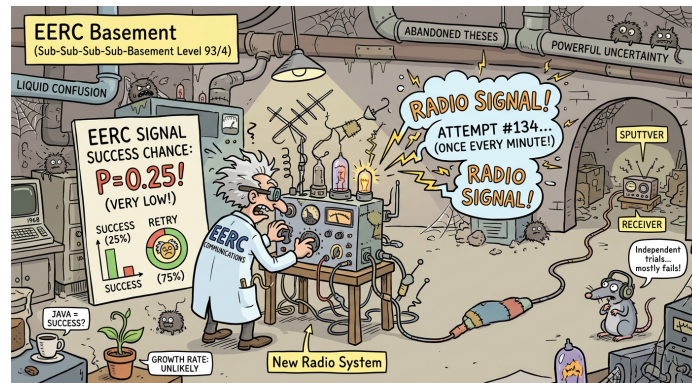
A batch contains $n = 5$ bulbs. Find the probability that **exactly 4 of the 5 bulbs** will still be working after 10,000 hours.

Example 7: Yoga Class

A yoga class in Yogavita Little Italy has 9 women and 6 men. If the class meets 10 times with perfect attendance at each meeting, and a student is selected at random at each meeting, what is the probability that a woman is selected 7 times (out of the 10 times)?



Example 8: Radio Communications System

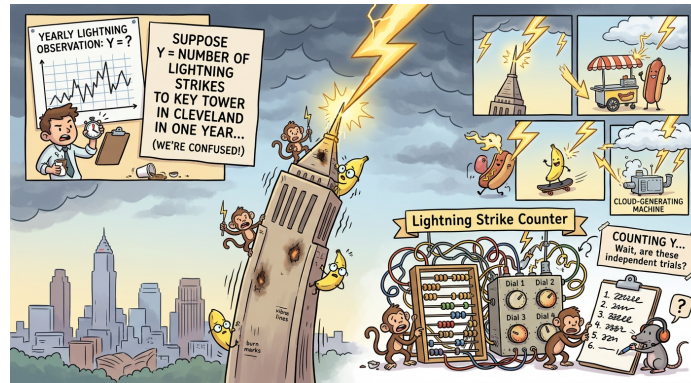


A new radio communications system attempts to send a signal to a receiver in the basement of the EERC once every minute. The probability of a successful transmission on each attempt is $p = 0.25$, and the attempts are independent.

Let Y = number of attempts required to observe the first successful transmission.

- (a) What kind of discrete random variable is Y ?
- (b) What is the probability that the **first** successful transmission occurs on the **fourth attempt**?
- (c) **On average**, how many attempts are needed to observe the first success?

Example 9: Lightning Strikes



Suppose, Y = number of lightning strikes to Key Tower in Cleveland in one year. Assume that $Y \sim \text{Poisson}(\lambda = 0.125)$, where $\lambda = 0.125 = \frac{1}{8}$.

- (a) Find $\mu = E(Y)$ and interpret the result in the context of lightning strikes.
- (b) Find the probability of observing **no lightning strikes during the next year**.
- (c) Find the probability of observing **at least one lightning strike within the next year**.